WEEK 1 · DAY 1 · RULE BOOK

Colab Bootcamp + Your First Time Series

Follow it step by step. Every common error and its fix is here.

Your goal today: open the exercise in Colab, load 3 years of Ka-band availability data, and earn **12 green ✅ ticks**. Time: about 60 minutes. Keep this page open next to Colab.

Before you start (2 minutes)

- ☐ Open **Google Chrome** and sign in to your Google account (top-right of Chrome).
- ☐ Click the **Open in Colab** button for `W1D1-exercise`.
- ☐ **File → Save a copy in Drive.** A new tab opens. **Work only in that new tab.**
- ☐ Run **STEP 0** (click it, press **Shift + Enter**). Wait for ✅ `Ready!`

The 7 golden rules

#	Rule	Why
1	**Save a copy in Drive first**	The original can't be saved. Your work would vanish.
2	**Run cells top to bottom**, one at a time	Each cell uses what the cells above created.
3	**Shift + Enter** runs a cell and jumps to the next	Fastest way to move through the notebook.
4	**Replace every** `___` with your answer, then run that cell	`___` is the blank. The check only marks filled-in answers.
5	**Text goes in quotes**: `"timestamp"`, `"August"`	Without quotes Python thinks it's a variable name → `NameError`.
6	**After any restart, run STEP 0 again**	A restart wipes Colab's memory.
7	**Never paste real NIGCOMSAT data into the AI (Gemini) button**	Course data is synthetic and safe. Real telemetry is restricted.

The exercise, step by step

Task	What you do	The tool you need
A1	Run the hello cell, then put your name in the message	Shift + Enter
A2	Hours in a year	`24 * 365`

Task	What you do	The tool you need
A3	Restart, see `NameError`, fix it	Runtime → Restart session, then STEP 0
A4–A5	Edit a text cell; find the 📁 Files panel	Double-click; left sidebar
B1	Is `timestamp` a date or text?	Read the `dtypes` output
B2	Name the time column	`"timestamp"` in quotes
B3	Count the days	`len(ka)`
B4	First and last day	`ka.index.min()`, `ka.index.max()`
B5	Reveal the 3 hidden missing days	frequency code for daily: `"D"`
B6	Find the worst day	`ka["availability_pct"].idxmin()`
B7	Chart title that states a finding	a sentence, in quotes
B8	Worst month on average	read the bar chart
B9	Trend, seasonality, cycle or noise?	the 4 characters (see glossary)
B10	21:00 UTC on a Lagos clock	WAT = UTC + 1
🌶️ **Bonus**	Quietest day for NOC tickets	copy the 3-line recipe

The 3-line recipe (you'll use it every day of this course)

```
df = pd.read_csv(data_path("FILE_NAME"),
                 parse_dates=["TIME_COLUMN"],   # 1. text → real dates
                 index_col="TIME_COLUMN")       # 2. time becomes the row label
df = df.asfreq("D")                             # 3. declare the rhythm: "h" hourly, "D"
daily, "W-MON" weekly, "MS" monthly
```

When things go wrong: error → fix

You see	It means	Fix
`NameError: name 'ka' is not defined`	Colab forgot (restart), or you skipped a cell	Run STEP 0, then the cells above, top to bottom
`NameError: name 'timestamp' is not defined`	You forgot the quotes	Write `"timestamp"`
`NameError: name 'check' is not defined`	STEP 0 was not run	Run STEP 0
`KeyError: 'timestamp'`	Wrong column name or spelling	Copy the name exactly from `head()`

You see	It means	Fix
`SyntaxError`	A typo: missing `)`, `"` or `,`	Compare with the recipe above, character by character
✍️ `Fill in the blank first`	A `___` is still there	Replace it, run again
Chart doesn't show	The cell didn't finish	Re-run it; make sure `plt.show()` is at the end
"This notebook was not authored by Google"	Normal warning	Click **Run anyway**
Everything is weird	–	**Runtime → Restart session and run all**

Stuck? Climb the ladder

1. **Read the ❌ hint**: it usually says exactly what's wrong.
2. **Check this rule book's error table.**
3. **Ask your neighbour** (explaining helps both of you).
4. **Ask the AI button** to *explain the error*. Paste your code and the error, never real data.
5. **Raise your hand.**

🏁 **Finished?** Run `scoreboard()`. 12 of 12 = 🏆. Then try the 🌶️ bonus and help someone near you.